Chapter 3 Digestion

Lesson 3-1:

Digestion and How Does It Work?





A. Complete the Following

Fill in the blanks with the correct words.



1. Digestion changes the food we eat into simple ___nutrients___ that the body can absorb.
2. Food moves from the mouth to the stomach through a tube called the ___esophagus___.
3. The stomach is located mainly in the ___upper___ abdomen.
4. In a cat, waste is stored in the ___large intestine___ before leaving the body.
5. In a fish, food passes from the mouth to the esophagus and then reaches the ___stomach___.



B. Write True or False

Write True (T) or False (F).



1. Teeth start breaking down food in the mouth. (T)
2. The small intestine is shorter than the stomach. (F)
3. The large intestine surrounds the small intestine in the human digestive system. (T)
4. Cats and fish both have digestive systems, although their digestive organs are arranged differently. (T)
5. All animals need food to live and grow. (T)

C. Cross Out the Odd One

In each group, one word does not belong. Cross it out.

1. Mouth – Esophagus – Stomach – ~~Trachea~~
2. Stomach – Small intestine – Large intestine – ~~Lungs~~
3. Chest – Upper abdomen – Middle abdomen – ~~Alveoli~~
4. Cat – Fish – Human – ~~Solar panel~~
5. Digestion – Absorption – Removal of waste – ~~Inhalation~~







Amazing Bodies Brighter Futures!

Chapter 3 Digestion



Learn Explore Grow!

Lesson 3-1:

Digestion and How Does It Work?

Healthy Food Healthy You!



D. Write the Scientific Term

Write the correct scientific term for each description.

Think Recall Write!

1	The process of breaking down food into smaller substances that the body can absorb.	Digestion
2	The organ that stores food and mixes it with strong digestive juices.	Stomach
3	The organ where most nutrients from digested food are absorbed into the blood.	Small intestine
4	The organ that absorbs water from leftover food and helps form waste.	Large intestine
5	The opening through which solid waste leaves the digestive system.	Anus

E. What Happens When...?

Write what happens in each situation.

Think Explain Be a Scientist!



1	Food enters the mouth?	Food is chewed and mixed with saliva to begin digestion.
2	Food reaches the stomach?	The stomach continues breaking down food by producing digestive juices.
3	Digested food reaches the small intestine?	Nutrients are absorbed into the blood through the walls of the small intestine.
4	Leftover food enters the large intestine?	Water is absorbed and the leftover waste becomes more solid.
5	Food passes through the digestive system of a fish and reaches the end of its intestine?	The food passes through the fish's intestine and waste leaves through the anus.

F. Give Reasons for the Following

Write the reason for each statement.

Think Deeper Give a Reason You Can Do it!

1	Food must be digested before the body can make full use of its nutrients.	Digestion breaks food into simple soluble substances that the body can absorb and use.
2	The stomach and intestines are important parts of the digestive system.	The stomach digests food, and the intestines complete digestion, absorb nutrients and absorb water/remove waste.
3	The small intestine is long and coiled inside the abdomen.	Its length and coils provide a large surface area and enough time for nutrient absorption.
4	The large intestine is important even though most nutrient absorption occurs in the small intestine.	It absorbs water and forms/stores waste, helping remove it from the body.
5	A cat and a fish can digest food even though their digestive systems are not arranged in exactly the same way.	They have the same basic digestive functions, but their differ in shape, size and arrangement.



Good Food Bright Minds Brighter Futures!



Chapter 3 Digestion

Lesson 3-2:

The Function of the Saliva





Try

Follow the Right Path for a Healthy You!

1 (1) What do we call the process of changing food into a form that is easy to absorb into the body?

Digestion

(2) Choose the correct words from below to fill in the blanks.

Mouth → Food pipe → Stomach → Small intestine → Large intestine → Anus

Large intestine / Stomach / Food pipe / Lungs / Small intestine

Exercise

Good Digestion Brighter Days!



1 (1) What is the correct order that food passes through? Choose the correct answer from below.

A: Mouth → Stomach → Food pipe → Small intestine → Large intestine
B: Mouth → Food pipe → Stomach → Large intestine → Small intestine
C ✓ Mouth → Food pipe → Stomach → Small intestine → Large intestine
D: Mouth → Food pipe → Large intestine → Small intestine → Stomach

(2) Alimentary canal

2 (1) In the experiment to check the function of saliva, why do you warm the test tubes in hot water of about 40 degrees Celsius? Choose the correct answer from below.

A: To disinfect.
B: To evaporate the water.
C ✓ To make it about the same temperature as inside the body.



(2) After mixing saliva and starch and warming it, did the color change when you added the iodine solution? Choose the correct answer from below.

It did not change. ✓



Healthy Habits Happier You!



Chapter 3 Digestion

Lesson 3-2:

The Function of the Saliva





A. Complete the Following

Fill in the blanks with the correct words.

1. The first step of digestion takes place in the mouth.
2. Saliva is a digestive found in the mouth.
3. Saliva helps break down starch in food.
4. Starch is changed into a simple sugar that can be easily absorbed by the body.
5. A liquid that helps digest food is called a digestive juice.



B. Write True or False

Write True (T) or False (F).

1. Digestion begins only after food reaches the stomach. (F)
2. Food begins to change from a complex form into a simpler form in the mouth. (T)
3. Saliva helps the body digest some of the food we eat. (T)
4. Digestive juices are produced only in the mouth. (F)
5. The stomach and intestines also produce digestive juices. (T)





Good Food
Bright Minds
Brighter Futures!



Chapter 3 Digestion

3-3

How Absorption Works



Explore Learn Grow!



Try

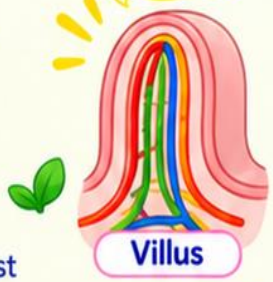
Tiny Bumps Big Help! Nutrients for a Stronger You!

1 (1) What do we call the small bumps on the inner wall of the small intestine that absorb nutrients?

villi

2 (2) What carries the absorbed nutrients to the body? Choose the correct answer from below.

- A: **Blood vessels**
- B: Nerves
- C: Muscles



3 (3) Where do the nutrients absorbed in the small intestine pass through first to be carried to the whole body? Choose the correct answer from below.

- A: Stomach
- B: Large intestine
- C: **Liver**

Healthy Food Better Absorption Happier You!

Exercise

1 (1) What is the advantage of having villi on the wall of the small intestine? Choose the correct answer from below.

- A: It makes it easier to swallow food.
- B: **The surface area of the wall of the small intestine becomes larger, and nutrients can be absorbed efficiently.**
- C: It makes it easier for nutrients to pass through.

A Larger Surface Area A Healthier You!



2 (2) Fill in the blanks with the correct words.

The nutrients of digested food, along with water, are mainly absorbed in the (① small intestine). The nutrients absorbed in the small intestine are carried by the (② blood vessels) through the (③ liver) to the whole body.



Good Digestion Brighter Minds Brighter Futures!



Chapter 3 Digestion

Lesson 3-3

How Absorption Works

Explore Learn Grow!

Our body absorbs digested nutrients and uses them for energy, growth and repair.

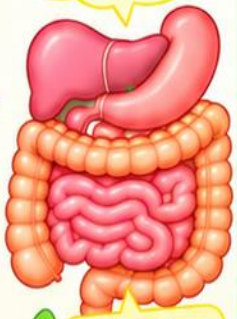


A. Choose the Correct Answer

Choose the correct answer.

Tiny Bumps Big Help! Nutrients for a Stronger You!

1. Which structure increases the surface area of the small intestine for absorption?
   a) Stomach (b) **Villi** c) Liver d) Esophagus
2. After nutrients are absorbed through the wall of the small intestine, they enter the:
   (a) **blood vessels** b) stomach c) large intestine. d) mouth.
3. Where do absorbed nutrients pass through before being carried to the rest of the body?
   a) Lungs (b) **Liver** c) Stomach d) Esophagus
4. Which substance produced by the liver helps break fats into tiny droplets?
   a) Saliva b) Water (c) **Bile** d) Blood
5. Which statement best describes one way the liver helps clean the body?
   a) It absorbs oxygen from the air. (b) **It breaks down harmful substances.**
   c) It stores solid waste. d) It pumps blood around the body.



B. Write True or False

Write True (T) or False (F).

A Healthy Liver Keeps You Healthy!

1. Absorption is the movement of digested nutrients and water from the intestine into the blood. (T)
2. Villi make the inner surface area of the small intestine smaller. (F)
3. The nutrients in food must become small and simple before they can be absorbed efficiently. (T)
4. The liver can store energy and release it when the body needs it. (T)
5. Blood vessels inside the villi carry absorbed nutrients away from the small intestine. (T)



C. Complete the Following

Fill in the blanks with the correct words.

More Surface Area More Nutrients for a Stronger You!

1. Most nutrients from digested food are absorbed in the **small** intestine.
2. The tiny bumps lining the inner wall of the small intestine are called **villi**.
3. Villi contain many **blood** vessels that carry absorbed nutrients.
4. The liver changes some nutrients from food into a simple form and stores them as **energy**.
5. Harmful substances broken down by the liver can be removed from the body through **urine**.



Good Nutrition Brighter Minds Brighter Futures!



Chapter 3 Digestion

Lesson 3–3

How Absorption Works

Explore Learn Grow!

Small Intestine Absorbs Nutrients!



Our body absorbs digested nutrients and uses them for energy, growth and repair.

D. What Happens When...?

Write what happens in each of the following situations.

Think Like a Scientist!



1. Digested nutrients reach the villi of the small intestine?
   They are absorbed through the villi into the blood vessels.
2. The inner wall of the small intestine has many villi instead of being smooth?
   The surface area becomes larger so more nutrients can be absorbed.
3. Absorbed nutrients enter the blood vessels inside the villi?
   They are carried in the blood to the whole body.
4. The body needs some of the energy that has been stored by the liver?
   The liver breaks down stored food (glycogen) and releases energy.
5. The villi in the small intestine are damaged or reduced in number?
   Less nutrients are absorbed, which can lead to poor growth and less energy.

Observe Think Explain Learn!

E. Give Reasons for the Following

Write the reason for each of the following.

Good Reasons Show Great Thinking!



1. The small intestine is an important organ for absorption.
   Because it has many villi which increase the surface area for absorbing nutrients.
2. Villi are useful for absorbing nutrients efficiently.
   Because they provide a large surface area for absorption.
3. The liver is important after nutrients have been absorbed into the blood.
   Because it stores energy and changes some nutrients into simpler forms.
4. The liver helps keep the body safe and healthy.
   Because it removes harmful substances from the body.
5. Breaking fats into tiny droplets helps the process of digestion.
   So digestive juices can easily break them and so they will be easily absorbed.





Healthy Digestion Brighter Minds Brighter Futures!